

Hao Yang

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📍 Washington, D.C.

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Education

Hong Kong University of Science and Technology , Electronic and Computer Engineering	Hong Kong 2022 – 2024
City University of Hong Kong , Electronic and Communication Engineering	Hong Kong 2018 – 2022

Experience

Hong Kong University of Science and Technology , Research Assistant Proposed a scalable framework for efficient FFT implementation on Versal ACAP, featuring an analytical QoR model and an automated code generator for rapid design deployment.	Hong Kong 2024 – 2025 1 year
City University of Hong Kong , Research Assistant Supported FPGA and embedded-systems projects by contributing to design, implementation, debugging, and performance optimization across hardware and software components.	Hong Kong May 2020 – Sept 2020 5 months

Volunteer

CityU EE Gifted Education Fund Programme 2021 , Tutor The programme was divided into phases, starting with AI/IoT fundamentals, followed by programming, and hands-on laboratory sessions to build CNN models.	Hong Kong Feb 2021 – Jan 2022
IEEE HK Section , Webmaster Maintained and updated the IEEE Hong Kong Section website, ensuring content accuracy and reliable site operation.	Hong Kong 2022 – 2025

Awards

HKSAR Government Scholarship Fund Awards scholarships to outstanding local and non-local full-time students in publicly funded sub-degree and above programs across UGC-funded universities and selected institutions, including the Scholarship for Outstanding Performance for non-local degree students (HK\$80,000 per year). HKSAR Government	2021
CityU Mainland Student Scholarship Scheme - Full Tuition Scholarship Entrance scholarships for outstanding Mainland students, including a Full Tuition Scholarship covering annual tuition fees (HK\$120,000 per year). City University of Hong Kong	2021
College of Engineering Dean's List City University of Hong Kong	2019
HKSAR Government Scholarship Fund HKSAR Government	2020
CityU Mainland Student Scholarship Scheme - Full Tuition Scholarship City University of Hong Kong	2020

HKSAR Government Scholarship Fund Awards scholarships to outstanding local and non-local full-time students in publicly funded sub-degree and above programs across UGC-funded universities and selected institutions, including the Scholarship for Outstanding Performance for non-local degree students (HK\$80,000 per year). HKSAR Government	2019
CityU Mainland Student Scholarship Scheme - Full Tuition Scholarship City University of Hong Kong	2019
CityU Mainland Student Scholarship Scheme - Full Tuition Scholarship City University of Hong Kong	2018

Publications

ESFA: An Efficient Scalable FFT Design Framework on Versal AI Engine

This paper proposes the ESFA framework, which can efficiently and automatically implement a scalable FFT on the Versal ACAP with multiple AIEs.

Hao Yang, Linfeng Du, Shangkun Li, Wei Zhang

ieeexplore.ieee.org/abstract/document/11173669

ESFA: An Efficient Scalable FFT Design Framework on Versal AI Engine

Hao Yang, Linfeng Du, Shangkun Li, Wei Zhang

Skills

FPGA

Programming

Machine Learning

Languages

Chinese

Native speaker

English

Fluent

Interests

Sports

Entertainment

Projects

Efficiently large point FFT Acceleration on Versal ACAP architecture

Sept 2022 – July 2024

Developed an accelerator on the Xilinx VCK190 platform to improve the throughput and efficiency of large-scale point FFT computations.

- FPGA

MIPS based 5-stage pipeline processor

June 2022 – Sept 2022

Implemented a Verilog-based MIPS-compatible processor featuring a five-stage pipelined architecture.

GAN Model Acceleration

Sept 2021 – Aug 2022

Proposed a GAN-based underwater image color-correction model and deployed it on the Atlas 200 DK board to enable real-time inference.

- GAN model